

RollNo. 

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### ODD END SEMESTER EXAMINATION DECEMBER – 2025

Name of the Course: B.Tech

Semester: 3

Name of the Paper: Electrical Engineering Materials

Paper Code: TEE-305

Time: 3 Hours

Maximum Marks: 100

**Note:-**

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

|           |                                                                                                  |     |
|-----------|--------------------------------------------------------------------------------------------------|-----|
| <b>Q1</b> | (20marks)                                                                                        |     |
| (a)       | Derive Bragg's law and explain its application in X-ray diffraction.                             | CO1 |
| (b)       | Classify different bonds in solid materials.                                                     |     |
| (c)       | Explain structural imperfections in crystals and their effects on material properties.           |     |
| <b>Q2</b> | (20marks)                                                                                        |     |
| (a)       | Differentiate between electrical conducting and insulating materials based on their properties.  | CO2 |
| (b)       | Describe in detail the optical behavior of metals, semiconductors, and insulators.               |     |
| (c)       | Explain superconducting materials and their properties.                                          |     |
| <b>Q3</b> | (20marks)                                                                                        |     |
| (a)       | Discuss the factors affecting the electrical conductivity of metals.                             | CO3 |
| (b)       | Explain the difference between intrinsic and extrinsic semiconductors with energy band diagrams. |     |
| (c)       | Explain drift and diffusion currents.                                                            |     |
| <b>Q4</b> | (20marks)                                                                                        |     |
| (a)       | Explain polarization mechanisms in dielectric materials with neat diagrams.                      | CO4 |
| (b)       | What is dielectric loss? Explain the factors affecting dielectric loss and dielectric strength.  |     |
| (c)       | Discuss dielectric properties of ferroelectric materials.                                        |     |
| <b>Q5</b> | (20marks)                                                                                        |     |
| (a)       | Explain the origin of permanent magnetic dipoles in matters.                                     | CO5 |
| (b)       | Differentiate between Diamagnetism, Paramagnetism, and Ferromagnetism with examples.             |     |
| (c)       | Write short notes on magnetostriction and its significance.                                      |     |

